

2. Develop a Fixed Deposit (FD) and Recurring Deposit (RD) Interest Calculator using Java Servlets to perform server-side financial computations based on user input through HTML forms.

Index.html

```
<!DOCTYPE html>

<html>

  <head>

    <title>Interest Calculator</title>

  </head>

  <body>

    <h1>FD/RD Interest Calculator</h1>

    <form action="interest" method="post">

      <select name="type">

        <option value="FD"> Fixed Deposit</option>

        <option value="RD"> Recurring Deposit</option>

      </select> <br> <br>

      Principal Amount: <input type="number" name="principal"><br> <br>

      ROI: <input type="number" name="roi"><br> <br>

      Time(in years): <input type="number" name="time"><br> <br>

      <button type="submit"> Calculate </button>

    </form>

  </body>

</html>
```

InterestCalculator.java

```
package com.og.practical2;
```

```
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.IOException;
import java.io.PrintWriter;
```

```
@WebServlet("/interest")
```

```
public class InterestCalculator extends HttpServlet {
```

```
    @Override
```

```
    protected void doPost(HttpServletRequest req, HttpServletResponse resp) throws
        ServletException, IOException {
```

```
        String type = req.getParameter("type");
        double p = Double.parseDouble(req.getParameter("principal"));
        double r = Double.parseDouble(req.getParameter("roi"));
        double t = Double.parseDouble(req.getParameter("time"));
```

```
        double interest;
```

```
        double amount;
```

```
        PrintWriter out = resp.getWriter();
```

```
        resp.setContentType("text/html");
```

```
        if(type.equals("FD")){
```

```
            interest = (p*r*t)/100;
```

```
    amount = p + interest;

    out.println("<h1> Your Total amount is: "+ amount + "</h1>");

    out.println("<h1> Your Interest is: "+ interest + "</h1>");
}
else{
    p = p*t*12;
    interest = (p*r*t)/100;
    amount = p + interest;

    out.println("<h1> Your Total amount is: "+ amount + "</h1>");

    out.println("<h1> Your Interest is: "+ interest + "</h1>");
}
}
}
```